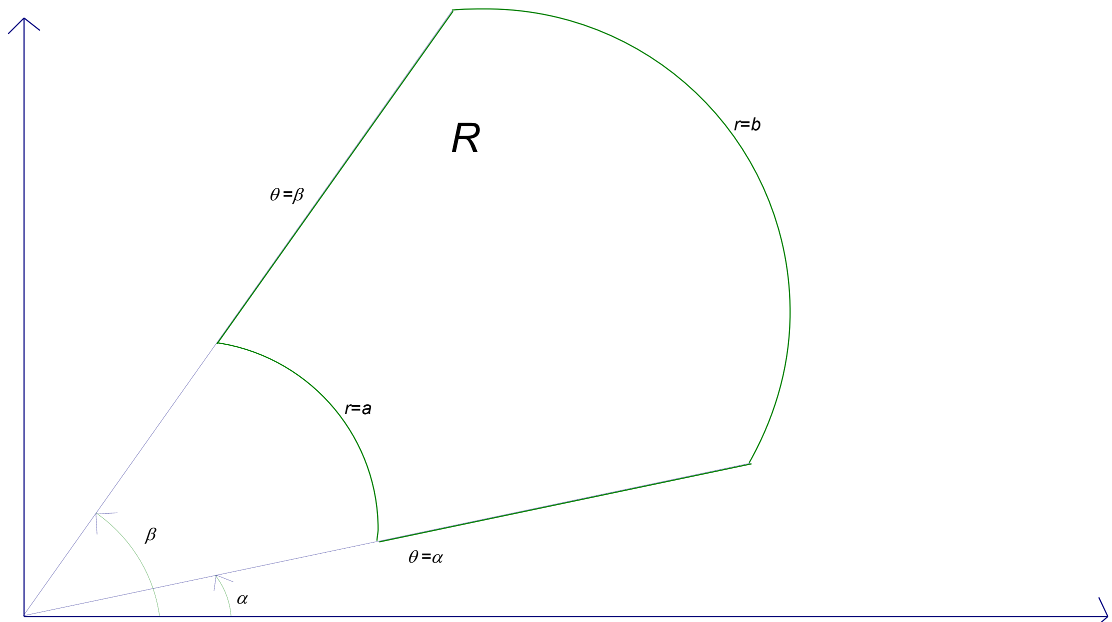
4.3 Double Integrals in Polar Coordinates

Recall: The identities that link the polar coordinates (r,θ) of a point to the rectangular coordinates (x,y) of the same point are $x = r\cos(\theta)$, $y = r\sin(\theta)$, and $r^2 = x^2 + y^2$.

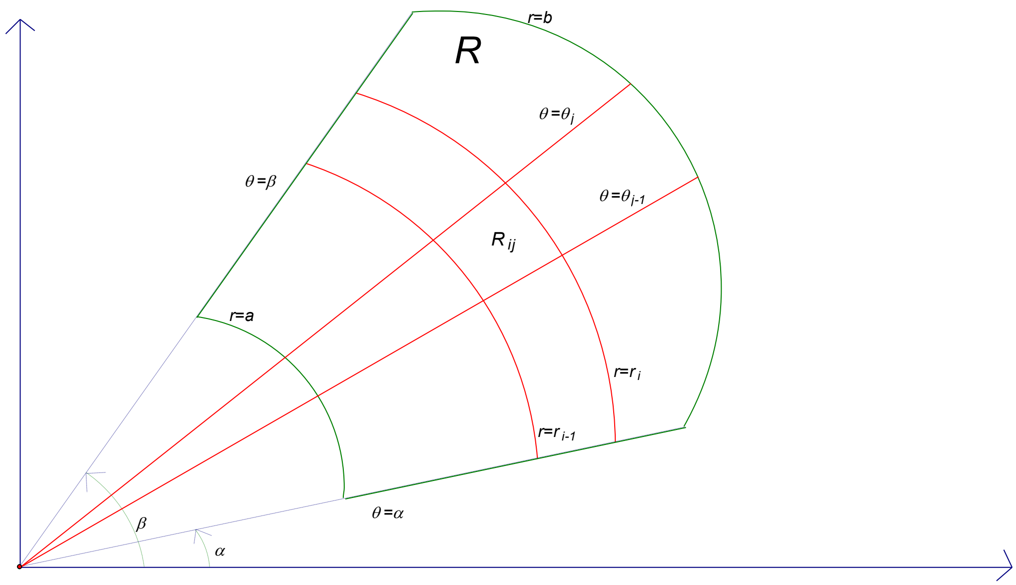
A <u>polar rectangle</u> is a region of the form $R = \{(r,\theta) \mid a \le r \le b,\ \alpha \le \theta \le \beta\}$. A picture of a polar rectangle is (the actual rectangle is outlined in green):



Motivating Question: In this section, our first concern will be trying to find a way to find the signed volume of a solid bounded by some function $f(x,y) = f(r\cos(\theta), r\sin(\theta))$ and a polar rectangle. Mathematically, we wish to compute $\iint_R f(x,y)\,dA$ where $R = \{(r,\theta) \mid a \le r \le b\ \alpha \le \theta \le \beta\}$ is a polar rectangle and f is continuous.

We know that $\iint_R f(x,y)\,dA$ is the signed volume of the solid bounded by the surface determined by $f(x,y)$ and the polar rectangle R. So, the idea to calculate the signed volume is to divide $[a,b]$ into m subintervals of the form $[r_{i-1}, r_i]$ each with width Δr and to divide the interval $[\alpha,\beta]$ into n subintervals of the form $[\theta_{j-1}, \theta_j]$ each with width $\Delta\theta$.

Then the circles $r = r_i$ and the rays $\theta = \theta_j$ divide the polar rectangle into smaller polar rectangles. A picture of such a rectangle is shown below:



In actuality the above picture would have many more red lines and many more small polar rectangles.

In order to approximate $\iint_R f(x,y)\,dA$ we will add up the volume of prisms which have bases that are polar rectangles and heights given by the function $f(x,y)$. The idea is that our approximation will approach perfection as both *m* and *n* approach infinity.

To find the volume of such prisms we need to find a formula for the area of R_{ij}. We see that the area ΔA_{ij} of R_{ij} is the difference of the area of two circle sectors. So, we have that

$$\Delta A_{ij} = \frac{\Delta\theta}{2\pi} r_i^2 \pi - \frac{\Delta\theta}{2\pi} r_{i-1}^2 \pi = \frac{1}{2}(r_i + r_{i-1})(r_i - r_{i-1})\Delta\theta = \frac{1}{2}(r_i + r_{i-1})\Delta r \Delta\theta.$$

We know we need to pick a sample point in R_{ij} to plug into $f(x,y)$ in order to get our height. So, we let $r_i^* = \frac{1}{2}(r_{i-1} + r_i)$ and $\theta_j^* = \frac{1}{2}(\theta_{j-1} + \theta_j)$. This means that our approximation becomes:

$$\iint_R f(x,y)\,dA \approx \sum_{i=1}^{m}\sum_{j=1}^{n} f\left(r_i^* \cos\left(\theta_j^*\right), r_i^* \sin\left(\theta_j^*\right)\right)\Delta A_{ij} = \sum_{i=1}^{m}\sum_{j=1}^{n} f\left(r_i^* \cos\left(\theta_j^*\right), r_i^* \sin\left(\theta_j^*\right)\right) r_i^* \Delta r \Delta\theta$$

As we let *m* and *n* approach infinity this approximation approaches perfection (one would have to prove that this idea that we used for rectangles also holds for polar rectangles). So, if we let $g(r,\theta) = f\left(r\cos(\theta), r\sin(\theta)\right) r$, we have that

$$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f\left(r_i^* \cos\left(\theta_j^*\right), r_i^* \sin\left(\theta_j^*\right)\right) r_i^* \Delta r \Delta\theta = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} g\left(r_i^*, \theta_j^*\right)\Delta r \Delta\theta.$$

We recognize the right most sum as a double integral! So, by Fubini's theorem

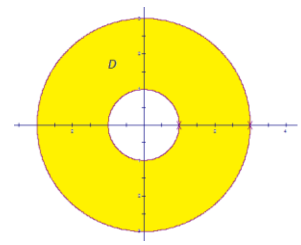
$$\lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} g\left(r_i^*, \theta_j^*\right)\Delta r \Delta\theta = \int_\alpha^\beta \int_a^b g(r,\theta)\,dr\,d\theta = \int_\alpha^\beta \int_a^b f\left(r\cos(\theta), r\sin(\theta)\right) r\,dr\,d\theta.$$

The following result summarizes the above derivation.

Change to Polar Coordinates in a Double Integral Over a Polar Rectangle):
If f is continuous on a polar rectangle R given by $0 \le a \le r \le b$, $\alpha \le \theta \le \beta$ where $0 \le \beta - \alpha \le 2\pi$, then
$$\iint_R f(x,y)\,dA = \int_\alpha^\beta \int_a^b f\big(r\cos(\theta), r\sin(\theta)\big)\, r\,dr\,d\theta.$$

Example 1: Find the volume of the solid that is under the hemisphere given by $z = \sqrt{16 - x^2 - y^2}$ and above the region bounded by $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$.

A picture of the region (call it *D*) is shown at the right.



We wish to find $\iint_D \sqrt{16 - x^2 - y^2}\,dA$. This integral appears to be difficult in rectangular (since *D* is not a type 1 or a type 2 region). So, we try to convert this integral to polar.

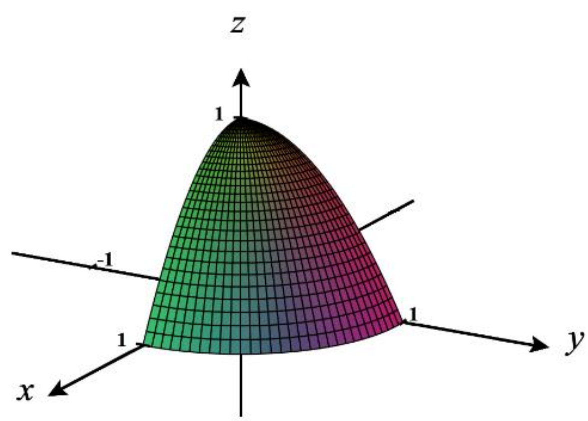
We note that in polar $D = \{(r,\theta) \mid 0 \le \theta \le 2\pi,\ 1 \le r \le 3\}$.

So, we have that $\iint_D \sqrt{16 - x^2 - y^2}\,dA = \int_0^{2\pi} \left[\int_1^3 \left(\sqrt{16 - r^2}\right) r\,dr \right] d\theta$.

For the innermost integral we let $u = 16 - r^2 \Rightarrow du = -2r\,dr \Rightarrow \frac{-1}{2}du = r\,dr$, and we note that $u(1) = 15$ and $u(3) = 7$. So,
$$\int_0^{2\pi} \left[\int_1^3 \left(\sqrt{16 - r^2}\right) r\,dr \right] d\theta = \int_0^{2\pi} \left[\frac{-1}{2} \int_{15}^{7} u^{1/2}\,du \right] d\theta = \int_0^{2\pi} \left[\frac{-1}{3} u^{3/2} \Big|_{15}^{7} \right] d\theta = \frac{1}{3}\int_0^{2\pi} \left(15\sqrt{15} - 7\sqrt{7}\right) d\theta = \frac{2\pi\left(15\sqrt{15} - 7\sqrt{7}\right)}{3}.$$

Example 2: Setup, but do not evaluate, an iterated integral which represents the volume of the solid bounded by the plane $z = 0$ and the surface $z = 1 - x^2 - y^2$ in the first octant.

Note that $z = 1 - x^2 - y^2$ intersects the *xy*-plane when $z = 0$.
When this happens, we see that $0 = 1 - x^2 - y^2 \quad \to \quad x^2 + y^2 = 1$, meaning the intersection is a circle of radius 1. Therefore, we recognize that we are integrating over the region
$$D = \left\{ (r,\theta) \mid 0 \le r \le 1\ \&\ 0 \le \theta \le \frac{\pi}{2} \right\}$$



Therefore, $V = \iint_D \left(1 - x^2 - y^2\right) dA = \int_0^{\pi/2} \int_0^1 \left(1 - r^2\right) r\,dr\,d\theta$.

One is able to generalize the above result, (with a method similar to how we generalized from rectangles to type 1 and type 2 regions in 4.2) and obtain the following theorem. If it is the case that *r* can be bounded by two functions rather than two constants we can update the pervious formula to the following.

Theorem (Change to Polar Coordinates in a Double Integral Over a Non-Rectangular Polar Region): If f is continuous on a polar region of the form $D=\{(r,\theta):\alpha\le\theta\le\beta,\ h_1(\theta)\le r\le h_2(\theta)\}$, then

$$\iint_D f(x,y)\,dA=\int_{\alpha}^{\beta}\int_{h_1(\theta)}^{h_2(\theta)} f\left(r\cos(\theta),r\sin(\theta)\right)r\,dr\,d\theta.$$

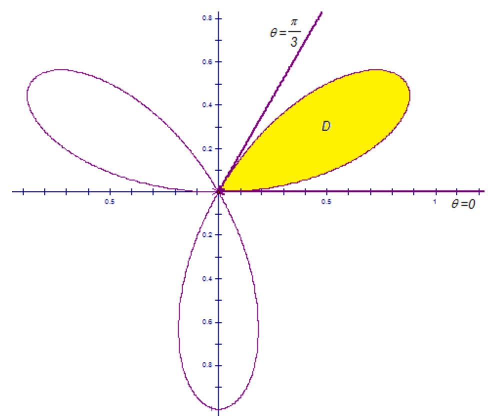
Note: A subtle point here is that two distinct points in D should not be in the same spot on the plane, if we are thinking of the double integral as giving a signed volume.

Note: $\iint_D dA$ *would yield the area of the polar region D.*

Note: It is possible to deduce the formula learned for polar integrals in calculus II from this theorem! Think about how this would be done.

Example 3: Use a double integral to find the area of one loop of the rose given by $r=\sin(3\theta)$.

A picture of the rose (with one shaded leaf) is:



We notice that the first leaf is traced as θ varies from 0 to pi over 3. Moreover, *r* stays nonnegative as the first leaf is traced. This means that our region of integration is $D=\left\{(r,\theta)\,|\,0\le\theta\le\frac{\pi}{3},\ 0\le r\le\sin(3\theta)\right\}$.

So, the area is

$$\iint_D dA=\int_0^{\frac{\pi}{3}}\left[\int_0^{\sin(3\theta)} r\,dr\right]d\theta=\int_0^{\frac{\pi}{3}}\left[\frac{1}{2}r^2\Big|_{r=0}^{r=\sin(3\theta)}\right]d\theta=\frac{1}{2}\int_0^{\frac{\pi}{3}}\sin^2(3\theta)\,d\theta$$

$$=\frac{1}{4}\int_0^{\frac{\pi}{3}}\left(1-\cos(6\theta)\right)d\theta=\frac{1}{4}\left(\theta-\frac{1}{6}\sin(6\theta)\right)\Big|_0^{\frac{\pi}{3}}=\frac{1}{4}\left(\frac{\pi}{3}\right)=\frac{\pi}{12}.$$

The next example is one of my most favorite math problems of all time. Those of you who may have taken me for Calculus II are aware of the awesomeness that is this next problem!

Example 4: Evaluate the following improper integral: $\int_0^\infty e^{-x^2}\,dx$.

This particular improper integral is quite difficult to evaluate because e^{-x^2} does not have an antiderivative which can be represented with elementary functions. However, in Calculus II we can prove that $\int_0^\infty e^{-x^2}\,dx$ converges (we just have no idea what it converges to).

So, suppose that we let $I = \int_0^\infty e^{-x^2}\,dx$. Then,

$$I^2 = \left(\int_0^\infty e^{-x^2}\,dx\right)\left(\int_0^\infty e^{-y^2}\,dy\right) = \int_0^\infty\left(\int_0^\infty e^{-x^2}\,dx\right)e^{-y^2}\,dy = \int_0^\infty\int_0^\infty e^{-x^2-y^2}\,dxdy = \int_0^\infty\int_0^\infty e^{-\left(x^2+y^2\right)}dxdy.$$

We note that the region of integration is the entire first quadrant.

In polar, we can represent the first quadrant as the set $D = \left\{(x,y)\,|\,0 \le \theta \le \frac{\pi}{2},\ r \ge 0\right\}$.

Now, we know that

$$\int_0^\infty\int_0^\infty e^{-\left(x^2+y^2\right)}dxdy = \int_0^{\frac{\pi}{2}}\int_0^\infty e^{-r^2}\,rdrd\theta = \int_0^{\frac{\pi}{2}}\left[\lim_{n\to\infty}\int_0^n e^{-r^2}\,rdr\right]d\theta = \int_0^{\frac{\pi}{2}}\left[\lim_{n\to\infty}\frac{-1}{2}e^{-r^2}\Big|_0^n\right]d\theta$$

$$= \int_0^{\frac{\pi}{2}}\left[\lim_{n\to\infty}\left(\frac{-1}{2}e^{-n^2}+\frac{1}{2}\right)\right]d\theta = \int_0^{\frac{\pi}{2}}\frac{1}{2}\,d\theta = \frac{\pi}{4}.$$

Thus, we have that $I^2 = \frac{\pi}{4} \Rightarrow I = \frac{\sqrt{\pi}}{2}$, and we know that $\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$!